Open Issues in `std::execution::task`

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- R1: March 2026 (post-Croydon mailing)
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Abstract

`std::execution::task` (P3552R3[1]) has open issues identified by national ballot comments, LWG issues, and published papers. This paper classifies each issue by whether it can be resolved after C++26 ships or whether shipping forecloses the fix.

Revision History

R1: March 2026 (post-Croydon mailing)

- Complete rewrite as an informational classification of open issues.

R0: February 2026 (pre-Croydon mailing)

- Original analysis of structural gaps. See P4007R0[2].

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1. Disclosure

The authors developed P4007R0[2] ("Senders and Coroutines"). The classification below holds regardless of whether any alternative design exists.

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2. Fixed After Ship

`task`'s `promise_type` is a class template instantiated in user code. Its `operator new`, allocator selection, stop token storage, environment forwarding, and destruction ordering can change between standard revisions without binary incompatibility. These issues are fixable post-ship: Unusual Allocator Customisation[3], Flexible Allocator Position[3], Shadowing The Environment Allocator[3], Stop Source Always Created[3], Stop Token Default Constructible[3], Task Not Actually Lazily Started[3], Frame Destroyed Too Late[3], No Default Arguments[3], `unhandled_stopped` Not `noexcept`[3], Environment Design Inefficient[3], Non-Sender Awaitables Unsupported[3], Future Language Feature Could Avoid `co_yield`[3], No TLS Capture/Restore Hook[3], `return_value` / `return_void` Have No Specification[3], `co_return { args... }` Unsupported[3], `change_coroutine_scheduler` Requires Assignable Scheduler[3], Sender-Unaware Coroutines Cannot `co_await` a `task`[3], Missing Rvalue Qualification[3], Parameter Lifetime Is Surprising[4], No Protection Against Dangling References[4], `co_yield with_error` Is Clunky[4], `co_await schedule(sch)` Is an Expensive No-Op[4], Coroutine Cancellation Is Ad-Hoc[4].

`affine_on` semantics, rescheduling behavior, and algorithm customisation are specification-level concerns. Tightening requirements or adding default implementations does not change any published interface. These issues are fixable post-ship: `affine_on` Default Implementation Lacks Specification[3], `affine_on` Semantics Not Clear[3], `affine_on` Shape May Not Be Correct[3], `affine_on` Shouldn't Forward Stop Requests[3],

`affine_on` Customisation For Other Senders[3], Starting a `task` Reschedules Unconditionally[3], Resuming After a `task` Reschedules Unnecessarily[3], `bulk` vs. `task_scheduler` [3], No Completion Scheduler[3], `with_awaitable_senders` Unused[3].

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3. Not Fixable Post-Ship

The issues in this section are items where shipping forecloses the fix.

Issue	References	Fixed
Allocator Timing	D3980R0[5], P3796R1[3], LWG 4356[6], US 254-385[7]	no
Allocator Propagation	D3980R0[5], P3796R1[3]	no
Error Return	P3950R0[8], P3801R0[4], P1713R0[9]	no
Symmetric Transfer	P2583R3[10], US 246-373[11], LWG 4348[12], P3801R0[4], P3796R1[3]	no

- **Allocator Timing.** `task` does not allow the user to specify an allocator at the launch site for coroutine frame allocation. The `allocator_arg` mechanism in P3552R3[1] requires the allocator to appear in the coroutine's parameter list, which creates a timing problem: the frame is allocated before the parameters are evaluated. Shipping this interface locks in the parameter-list approach and forecloses alternative injection mechanisms.

- **Allocator Propagation.** `task` does not propagate the frame allocator to child tasks. Each child task allocates its frame independently. Shipping without propagation locks in a design where every launch site must specify its allocator explicitly, foreclosing transparent propagation through the coroutine call tree.

- **Error Return.** `task` requires `co_yield with_error(e)` to propagate an error to the caller. `co_return` cannot carry an error value because `return_value` and `return_void` are mutually exclusive in the current coroutine specification. Shipping this interface locks in the `co_yield` mechanism and forecloses `co_return`-based error propagation, which would require a language change.

- **Symmetric Transfer.** `sender-awaitable::await_suspend` returns `void`. When a sender completes synchronously inside `start()`, the receiver calls `handle.resume()` on the `await_suspend` stack, adding a frame per completion with no upper bound. Shipping this interface locks in the `void` return type and forecloses `coroutine_handle<>` return, which would enable symmetric transfer.

Acknowledgements

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References

Papers, issues, and ballot comments referenced in this document.

WG21 Papers

1. P3552R3 - ''Add a Coroutine Task Type'' (Dietmar Kühl, Maikel Nadolski, 2025). https://wg21.link/p3552r3
2. P4007R0 - ''Senders and Coroutines'' (Vinnie Falco, Mungo Gill, 2026). https://wg21.link/p4007r0
3. P3796R1 - ''Coroutine Task Issues'' (Dietmar Kühl, 2025). https://wg21.link/p3796r1
4. P3801R0 - ''Concerns about the design of std::execution::task'' (Jonathan Müller, 2025). https://wg21.link/p3801r0
5. D3980R0 - ''Task's Allocator Use'' (Dietmar Kühl, 2026). https://isocpp.org/files/papers/D3980R0.html
6. LWG 4356 - ''connect() should use get_allocator(get_env(rcvr))''. https://cplusplus.github.io/LWG/issue4356
7. US 254-385 - C++26 NB ballot comment. https://github.com/cplusplus/nbballot/issues/960
8. P3950R0 - ''return_value & return_void Are Not Mutually Exclusive'' (Robert Leahy, 2025). https://wg21.link/p3950r0
9. P1713R0 - ''Allowing both co_return; and co_return value; in the same coroutine'' (Lewis Baker, 2019). https://wg21.link/p1713r0
10. P2583R3 - ''Symmetric Transfer and Sender Composition'' (Mungo Gill, Vinnie Falco, 2026). https://wg21.link/p2583r3
11. US 246-373 - C++26 NB ballot comment. https://github.com/cplusplus/nbballot/issues/948
12. LWG 4348 - ''task doesn't support symmetric transfer''. https://cplusplus.github.io/LWG/issue4348

Other

13. P2300R10 - ''std::execution'' (Michał Dominiak et al., 2024). https://wg21.link/p2300r10
14. C++ Working Draft - (Richard Smith, ed.). https://eel.is/c++draft/